KIRAN RAJEEV

ELECTRONICS AND COMMUNICATION ENGINEER



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LinkedIn
Portfolio

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in embedded systems, industrial networking and hardware-software integration. I am familiar with C, embedded systems fundamentals, ESP32-based IoT applications, TCP/IP, Modbus communication, and basic networking concepts. Currently developing technical skills in embedded systems and Linux, with an interest in hardware–software integration, embedded development, and network engineering. I am interested in working with a growth-oriented organization where I can apply my technical knowledge, contribute to research and product development, gain practical experience, and grow professionally

EDUCATION

Government Engineering College, Palakkad **2021 – 2025**
Bachelor of Technology in Electronics and Communication Engineering — CGPA: 8.6 *Palakkad, Kerala, India*

EXPERIENCE

SMEC Automation **Nov 2025 – May 2026**
CRG Trainee Engineer *Cochin, India*

- Configured and tested industrial communication over Modbus TCP/IP between PLC, HMI and SCADA systems, verifying the accurate data exchange between devices.
- Performed system troubleshooting, validation and functional testing during testing and commissioning activities.
- Integrated PLC controllers with HMI/SCADA software for real-time monitoring.
- Assisted in PLC, SCADA and HMI development, configuration and testing for industrial automation systems.

TECHNICAL SKILLS

Networking	TCP/IP,IP Addressing,Routing & Switching Fundamentals, Subnetting, DHCP, DNS,Basic Network Troubleshooting,Basic firewall concept & VLAN.
Industrial Communication	Modbus TCP/IP, Modbus RTU, Ethernet/IP, ASCII, RS-232, RS-485, UART, I2C, SPI
Programming	C/C++(learning), Python, Ladder Logic
Tools	Arduino IDE, Arduino IoT Cloud, Microsoft Office, Linux(learning), TIA Portal, WinCC Unified
Automation	PLC Programming, HMI Designing, VFD Parameter Configuration, Troubleshooting, Testing & Commissioning
PLC Platforms	Siemens, Allen-Bradley, Schneider, ABB, Mitsubishi, Delta, Omron
SCADA / HMI	Wonderware InTouch, Citect SCADA, WinCC, Vijeo Designer, Delta DOPSoft

PROJECTS

Smart Fuel System Using IoT *IoT / QR Payment*

- Developed an IoT-based smart fuel system enabling real-time communication between users and petrol pump operations.
- Implemented a QR-based payment interface to improve transaction transparency and reduce customer waiting time.
- Focused on improving system security, user convenience and interaction between the user and fuel dispensing system.

Electronic Medicine Dispenser *ESP32 / C/C++ / Arduino IoT Cloud*

- Designed and developed an automated medicine dispenser using ESP32 for scheduled dispensing of solid pills and liquid dosages.
- Integrated Arduino IoT Cloud and Wi-Fi for remote medication monitoring, user alerts and real-time telemetry.
- Implemented compliance tracking using IR sensors and piezo buzzers to verify intake and trigger missed-dose alerts.
- Interfaced servo motors, water pump motor and YF-S201 flow-rate sensor using C/C++ in Arduino IDE.

Image Denoising Using Machine Learning *Python / TensorFlow / OpenCV / PyTorch*

- Developed machine learning-based image denoising solutions for Gaussian, Salt & Pepper, Speckle and Uniform noise.
- Built an end-to-end image denoising web application using Django and OpenCV, enabling noisy image upload and side-by-side restoration results.
- Designed a CNN-based noise classification pipeline to classify five noise types: Gaussian, Salt & Pepper, Poisson, Speckle and Uniform.
- Implemented a Patch-Based U-Net architecture using TensorFlow/Keras for high-resolution image restoration while retaining spatial features and edges.
- Integrated automated quality metrics including PSNR, SSIM, FSIM and NPS for quantitative evaluation, achieving up to 16.61 dB PSNR improvement.

Controller-to-Controller Communication *Schneider M340 PAC / Modicon M221*

- Configured communication between Schneider PAC M340 and Modicon M221 PLC using READ_VAR and WRITE_VAR function blocks.
- Configured PAC M340 as the Master and Modicon M221 as the Slave controller.
- Established Master-Slave communication over Modbus TCP/IP for the controller to data exchange.

Automatic Mains Failure (AMF) *Allen-Bradley MicroLogix 1400 PLC*

- Designed and developed a miniature Automatic Mains Failure system using an Allen-Bradley MicroLogix 1400 PLC.
- Implemented relay logic and 24V DC switching for automated mains/generator control.
- Used a bus coupler for load sharing and integrated control components for reliable switching.
- Interfaced the system with Wonderware InTouch SCADA for real-time monitoring and data logging.

Batching Process Automation *Siemens S7-1200 PLC / TIA Portal / WinCC Unified*

- Developed an automated batching process using a Siemens S7-1200 DC/DC/DC PLC and ladder logic in TIA Portal.
- Implemented weight-based batching of city water, QR and KM products into the main mixing tank according to predefined proportions.
- Programmed a 3-minute agitator operation after achieving the required batch proportion before transferring material to filling lines.
- Designed WinCC Unified SCADA for real-time monitoring, process visualization, alarm indication and data logging.

CERTIFICATIONS

- **Embedded Systems - Ongoing**
- **Computer Architecture** – NPTEL, IIT Madras
- **Embedded Systems in EV** – Techmaghi
- **Introduction to Telecommunications: Standards, Networks, and Innovations**-Linkdedin Learning
- **Professional Diploma in Industrial Automation (PDIA)** – IPCS Global — PLC Programming, SCADA, HMI, VFD, DCS
- **CSRBOX Placement Assistant Program** – IBM SkillsBuild

LEADERSHIP & VOLUNTEERING

- **Placement Coordinator**, ECE Department, Government Engineering College, Palakkad, India
- **Department Program Coordinator**, ECE Department, Government Engineering College, Palakkad, India